

Why invest in R&D in India: Experts' views

"With the intellectual capital available in the country and competitive talent costs, the research ecosystem in India offers a significant opportunity for global companies to establish their R&D centres in India, which can ultimately drive domestic manufacturing."

—**Prakash Mallya, vice president and managing director, sales and marketing group, Intel India**

"India is a highly cost-effective R&D destination, and also has a high-quality, young workforce, as evidenced by major international brands investing in R&D or design facilities in India. Despite a nine per cent increase in operating costs of R&D centres in India, it is still 25 per cent cheaper than in China. If capitalised well, this can lead to ten to fifteen per cent growth in R&D in the country in the next couple of years."

—**Nitin Kunkolienker, president, Manufacturers' Association of Information Technology (MAIT)**

"Opportunities at hand in the Indian market are immense, coupled with strong and attractive investments and a supportive policy environment, which make India an ideal investment destination for R&D. While within the sector, verticals of advanced analytics and AI are expected to continue to take the lead, the electronics R&D sector will only be strengthened with the advent of newer technologies of the fourth industrial revolution. India is raising its game to become a research and design powerhouse. With research arms of global companies and specialised providers expanding, the electronics sector is expected to grow at a CAGR of ~24 per cent to US\$ 400 billion by 2020 as compared to US\$ 69 billion in 2012."

—**Rohit Madhok, vice president - engineering, GlobalLogic**

"The R&D sector in India has undergone a tremendous shift owing to growing market demands, higher investments and policy support from the government. The growing IT sector, development in infrastructure and availability of latest technology for manufacturing purposes as well as skilled labour make India the perfect venue for new R&D initiatives."

—**Ravi Raj, director - sales and support, NetRack**

ties to create and innovate value for people.

India has become a key contributor to the global R&D sector. Intended reduction in government's import bill is likely to further boost domestic electronics manufacturing. Demand for competitive technology will accelerate more investments from big companies.

Further, technology transitions such as rollout of 5G/LTE networks and the IoT will drive accelerated adoption of electronic products. This further augments investments in production, distribution and R&D in the forthcoming years. Now, the time is ripe to focus on R&D, to come up with sustainable solutions custom-made for the Indian market.

Indian government's role and initiatives

Investments currently being made in R&D have been mainly driven by the Central government, with the private sector and state governments found to be lagging. This is in contrast to advanced countries where the private sector is the dominant and driving force of R&D spending.

Government research is being driven by Central Public Sector

Enterprises (CPSEs) with majority of research being done by the power and petroleum sector. Ratan Watal, member secretary to Prime Minister's Economic Advisory Council (PMEAC), has said that growth in R&D expenditure should be commensurate with growth of GDP.

The government is providing the following incentives for R&D based investments:

- Tax-friendly Special Economic Zone policy for the next twenty years
- Incentives for R&D investments in technology, talent development and reskilling
- Encouraging funding for technology startups
- Easing taxation regime for the digital economy and promoting ease of doing business

To encourage investments in R&D, the government of India offers various tax incentives to businesses for carrying out R&D activities. These include incentives on revenue and capital expenditures incurred for carrying out scientific research. Some of these incentives, according to India Briefing, are given below:

- A concessional tax rate of ten per cent (plus applicable surcharge and tax) on income earned by royalty

in respect of patents developed and registered in India.

- A deduction of 150 per cent for manufacturing companies on in-house R&D expenditure, including capital expenditures on scientific research related to production of articles and products, filing patent applications and obtaining approvals from regulatory authorities.

- Custom duty exemption to the in-house unit of industries for consumables and equipment needed for R&D.

- GST concession for research institutions.

India's R&D expenditure ecosystem

In the Union budget of 2019-20, the government announced a massive investment of US\$ 841 million for Ministry of Science and Technology, US\$ 369.15 million for Department of Biotechnology and US\$ 700.38 million for Council for Scientific and Industrial Research.

PMEAC has released a report titled 'R&D Expenditure Ecosystem,' which states that **growth in R&D expenditure should be commensurate with the economy's growth and should be targeted to reach at least two per cent of GDP by 2022. It has remained constant at around 0.6 to 0.7 per cent of GDP,** and this is well below major countries such as the US (2.8 per cent), China (2.1 per cent), Israel (4.3 per cent) and Korea (4.2 per cent).

It has been suggested that a minimum percentage of turnover of the company must be invested in R&D by medium and large enterprises registered in India. This will stimulate the private sector's investment in R&D, from the current 0.35 per cent of GDP. The report also pitched for creating thirty dedicated R&D export hubs in national interest.

As per a NASSCOM report, India's engineering and R&D market has grown to US\$ 30 billion in the financial year 2019, and is estimated to touch US\$ 42 billion by 2022. And, as per an Indian Science and R&D Industry Report, it is speculated that the medical technology sector in India would reach US\$ 9.6 billion by the end of 2022, and US\$ 7.8 billion by 2020 from US\$ 5.7 billion back in 2017. **EFY**